

Limb Pain

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Pain may affect all or part of a leg or arm. [Pain in the joints](#) is discussed elsewhere.

Limb pain may be constant or occur irregularly. Pain may be precipitated by motion or have no relation to movement. Other symptoms, such as warmth, redness or other discoloration, numbness, or tingling, may also be present, depending on the cause of the limb pain.

Causes of Limb Pain

Injuries and overuse are the most common causes of pain in a limb, but people usually know when these events are the cause of their pain. This discussion covers limb pain unrelated to injury or strain. Among the disorders that cause such limb pain, most affect the legs more than the arms. There are many causes.

The **most common** causes are the following:

- Blood clot in a deep-lying vein ([deep vein thrombosis](#))
- Bacterial infection of the skin ([cellulitis](#))
- [Pressure on a spinal nerve root](#)

Cellulitis Due to Bacterial Infection



The leg and foot are swollen and red. A pen can be used to mark the area of cellulitis, so doctors can determine whether the condition is improving (receding) or worsening (spreading).

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Uncommon but serious causes that require immediate evaluation and treatment include

- Sudden blockage of an artery in the limb (acute [arterial occlusion](#))
- Deep soft-tissue infection
- [Heart attack](#) (arm pain only)

Other less common causes include bone tumors, bone infections ([osteomyelitis](#)), and nerve problems such as pressure on nerves or degeneration of nerves (such as caused by [diabetes](#) or long-term [alcohol use disorder](#)).

Evaluation of Limb Pain

It is particularly important to make sure the person does not have a sudden blockage of an artery because the limb can develop gangrene (death of tissue) if there is no blood flow for more than a few hours. The following information can help people decide when a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with limb pain, certain symptoms and characteristics are cause for concern. They include

- Sudden, severe pain
- Limb that is cold to the touch or pale
- Chest pain, sweating, shortness of breath, or palpitations
- Signs of severe illness (for example, confusion, fever, or collapse)
- Limb that is suddenly swollen, blistered, or has black spots
- Risk factors for [deep vein thrombosis](#), such as recent surgery, bed rest, or a cast on a leg
- New nerve deficits, such as weakness or numbness of the affected limb

When to see a doctor

People who have warning signs should see a doctor right away. People without warning signs should call a doctor. The doctor will decide how quickly the person needs to be seen based on the symptoms, age, and presence of other medical disorders.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the limb pain and the tests that may need to be done.

Doctors ask

- How long limb pain has been present
- Whether pain occurs at certain times or during specific activities
- How intense the pain is
- Whether the pain is sharp or throbbing
- Where the pain is located
- What activities trigger or worsen pain
- What the person does to relieve pain
- What other symptoms (such as numbness or tingling) occur along with the pain

Doctors look for symptoms that may indicate a cause of the pain. Some obvious findings may be very helpful in diagnosing the cause of limb pain. For example, back or neck pain suggests that a [nerve root](#) may be affected and fever suggests that the person has an infection. Shortness of breath and a rapid heart rate suggest blockage of an artery by a blood clot that has traveled from a leg to the lungs ([pulmonary embolism](#)). An irregular pulse suggests that the person may have a certain abnormal heart rhythm ([atrial fibrillation](#)) that has caused a blood clot to travel from the heart to block an artery in the leg.

The painful limb is inspected for color, swelling, and any skin or hair changes. The doctor also checks for pulses, temperature, tenderness, and crepitation (a subtle crackling sensation indicating gas in the soft tissue caused by a serious infection). Strength, sensation, and reflexes are compared between affected and unaffected sides. Blood pressure is sometimes measured in the ankle or wrist of the affected limb and compared with the blood pressure in an unaffected arm or leg. If blood pressure is much lower in the painful limb, it is likely that the arteries in the limb are blocked.



Some Causes and Features of Limb Pain

Cause*	Common Features†	Tests
Sudden, severe pain that develops within a few minutes		
Blockage of an artery in a limb, usually a leg, by a blood clot	Sudden, severe pain	Arteriography done immediately
	Coolness and paleness of the limb	
	After several hours, signs of nerve malfunction, such as weakness, numbness, tingling, or cramping	
	Weak or no pulse felt in the limb	
Sudden herniation of a disk in the spine	Pain and sometimes numbness that occurs in a line down the limb	Usually MRI
	Pain that is often worsened by movement	
	Often neck or back pain	
	Sometimes weakness in part of the affected limb	
<u>Heart attack</u> (myocardial infarction)	Pain in an arm, not a leg	ECG
	Sometimes pain or pressure in the chest or jaw	Blood tests for substances that indicate heart damage (cardiac biomarkers)
	Sometimes nausea, sweating, and shortness of breath	Sometimes angiography of the arteries of the heart



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